Ultrawide-Bandgap Aluminum Nitride Diodes for High-Frequency Radio Applications

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Research Question
Can simulation-guided redesign turn an aluminum nitride (AlN) Schottky diode built for power electronics into one fast enough for radio-frequency (RF) use?

Results at a Glance

Cutoff frequency	Junction capacitance	Breakdown	fc × breakdown
0.42 → 42.7 GHz	2 pF → 2 fF	≥ 38 V	~3×
forward conduction; ~100× vs the lab's diode	~1000× smaller junction	no breakdown observed; ~2× the GaN SBD [2]	vs published GaN and GaAs RF diodes

UWBG Semiconductors for RF

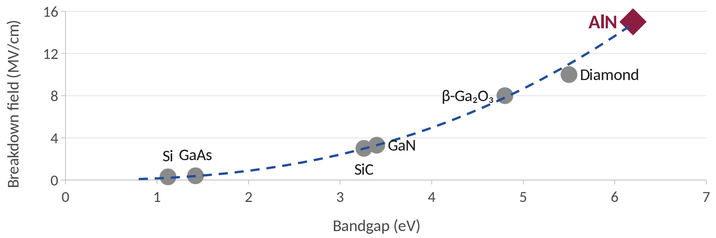


Fig. 1 Breakdown field vs bandgap; dashed line: $F_{BR} \propto E_G^{2.5}$.

- **UWBG advantage:** higher breakdown fields and lower leakage than Si, GaAs or GaN.
- **AlN:** 6.2 eV bandgap, 12–15 MV/cm critical field and ~5× GaN's Johnson's Figure of Merit (JFOM).

$$JFOM = V_{BR} \times f_T = F_{BR} v_{sat} / 2\pi$$

Motivation

- **Need:** 5G/6G and satellite links need diodes that are both fast and high-power.
- **Gap:** AlN Schottky diodes are built for power switching; the lab's device reaches only 0.42 GHz.

Methodology

- **Baseline:** the lab's AlN/AlGaN diode in Silvaco TCAD on ASU's Sol.
- **Calibration:** matched to measured data (resistance within ~25%).
- **Redesign:** AlN thickness and doping, anode shape and size.
- **Evaluation:** fc from DC, AC and charge; breakdown sweep.
- **Stress tests:** donor activation, contacts, field plates.

$$f_c = 1 / (2\pi R_s C_{j0})$$

Device Design

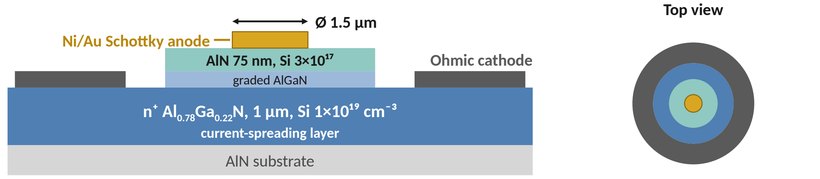


Fig. 2 (a) Cross-section and (b) top view of the final AlN diode.

- **Fully depleted 75 nm AlN:** acts as the capacitor's dielectric, so Cj ≈ 2 fF.
- **1.5 µm circular anode:** keeps the junction area small.
- **n⁺ AlGaN layer:** carries current sideways to a ring cathode.

Results and Discussion

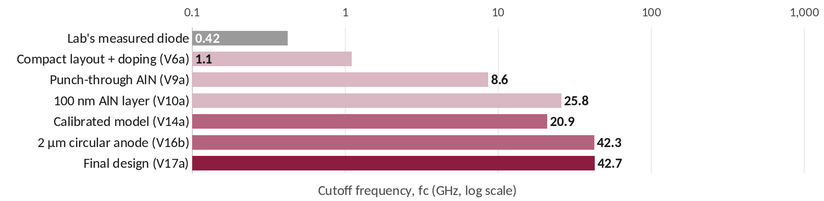


Fig. 3 Forward fc rose ~100× (0.42 → 42.7 GHz) across design steps; darker bars: calibrated model.

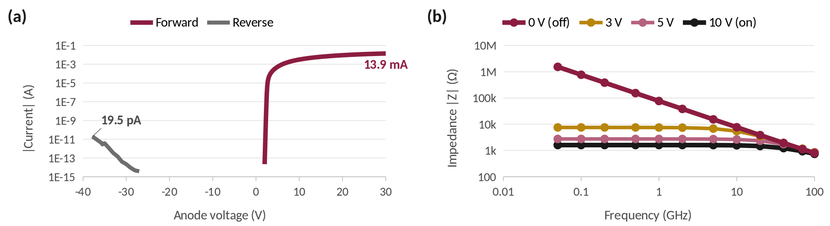
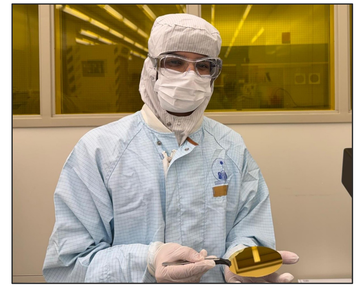


Fig. 4 (a) I–V: no breakdown to −37.8 V (19.5 pA). (b) Impedance: capacitive off, ~1.6 kΩ on.

Proposed Fabrication Flow



Proposed process flow

1. Mesa etch down to the AlGaN layer
2. Ti/Al/Ti ring cathode + anneal
3. Ni/Au anode (Ø 1.5 µm) by lift-off
4. SiO_2 deposition
5. Air bridge and probe pads
6. HF removal of the SiO_2 underneath

Air bridge links the 1.5 µm anode to large probe pads with little added capacitance.

Fig. 5 (a) Cleanroom training at the ASU NanoFab. (b) Proposed process flow.

Benchmark

Device	fc (0 V)	Breakdown	fc × V
GaAs SBD [1]	1.3 THz	10 V	13 THz·V
GaN SBD [2]	0.59 THz	18 V	11 THz·V
This work: AlN (simulated)	**0.95 THz**	**≥ 38 V**	**≥ 36 THz·V**

Table 1 Zero-bias fc and breakdown of published RF diodes [1], [2]; this work is simulated and intrinsic.

Conclusion

- **Feasibility:** thin, fully depleted AlN with a small anode raises fc ~100×.
- **Benchmark:** ~3× the fc × breakdown of published GaN and GaAs RF diodes.
- **Robustness:** forward fc within ~3%, Cj within ~5% at 10% activation.
- **Future work:** fabricate and measure I–V, C–V and RF; pads and transit time will lower measured fc.

Acknowledgements

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[1] B. Zhang et al., "A 220 GHz Balanced Frequency Doubler Based on GaAs Schottky Varactor Diodes," J. Electron. Mater., vol. 48, 2019, doi:10.1007/s11664-019-07067-z. [2] "120 GHz Frequency-Doubler Module Based on GaN Schottky Barrier Diode," Micromachines, vol. 13, 1172, 2022, doi:10.3390/mi13081172.

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